

Button Update Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

Immeas

BengalCatBalu

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Button on Button Update. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated February 13, 2026, was delivered to Button, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Button Update code provided by Button. The findings consist of 1 Medium & 5 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Button Update
Repository	button-protocol
Commit	15bb95edd97a...
Fix Commit	975b70442cd1...
Audit Timeline	Feb 2nd - Feb 6th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	1	1	0
Low Risk	5	4	1
Informational	7	7	0
Gas Optimizations	1	1	0
Total	14	13	1

6 Audit Scope

The audit scope was limited to:

```
script/AddApiWallet.s.sol
script/AddAssetToBasisTradeTailor.s.sol
script/AddToCreationWhitelist.s.sol
script/ApproveAgentInBasisTradeTailor.s.sol
script/ApproveEngineInBasisTradeTailor.s.sol
script/ApproveTailorInFactory.s.sol
script/ApproveWithdrawalDestination.s.sol
script/DeployMorphoBlueAdapter.s.sol
script/DeployPocketFactory.s.sol
script/GrantEngineRoleInBasisTradeTailor.s.sol
script/GrantOperatorRoleInBasisTradeTailor.s.sol
script/GrantOperatorRoleInPocketFactory.s.sol
script/ProcessWithdrawal.s.sol
script/RegisterAdapterInBasisTradeTailor.s.sol
script/RemoveAdapterFromBasisTradeTailor.s.sol
script/RevokeWithdrawalDestination.s.sol
script/TransferAssetFromCore.s.sol
script/TransferAssetToCore.s.sol
script/TransferPerp.s.sol
script/UpgradeBasisTradeTailor.s.sol
src/BasisTradeTailor.sol
src/Pocket.sol
src/PocketFactory.sol
src/adapters/BaseAdapter.sol
src/adapters/MorphoBlueAdapter.sol
src/libraries/CoreWriterEncoder.sol
```

Commit c352db7, made during the audit, and the post-audit commits 8d40ba7, 89b6c6c and 975b704 were also reviewed.

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Button/BasisTrade is an asset router designed to bridge between the EVM and HyperLiquid HyperCore as well as Morpho vaults. A per-user `Pocket` wallet is created via `BasisTradeTailor` and `PocketFactory`. The "pocket user" then deposits the configured `baseAsset` (e.g., USDC) directly into their Pocket through `BasisTradeTailor.deposit`. Withdrawals are two-step: the pocket user submits a withdrawal request (`requestWithdrawal`), which is later fulfilled by transferring the base asset out of the Pocket to the pocket user (`processWithdrawal`).

For trading and account management, the Tailor executes HyperCore actions from each Pocket by calling the CoreWriter system contract (`sendRawAction`), with actions encoded via `CoreWriterEncoder`. This includes adding API wallets, moving spot assets, transferring USD-class balances between spot/perp, and managing native HYPE staking flows (deposit/withdraw/delegate/undelegate and withdrawing HYPE).

Separately, the Tailor supports a protocol adapter execution path: registered adapters return a list of `Calls` (either External calls executed via `pocket.execWithValue` or Approve calls executed via `pocket.approve`). This enables

integrations such as Morpho Blue via the `MorphoBlueAdapter`. In the Morpho integration, the adapter builds calls to the Morpho core contract (e.g., borrow/repay, collateral operations) and includes prerequisite ERC20 approvals, allowing the Pocket to open and manage Morpho positions.

8.1 Key Components

- **BasisTradeTailor:** Central coordinator that (i) creates Pockets via `PocketFactory`, (ii) maps each Pocket to a `pocketUser`, (iii) supports user deposit + withdrawal request/processing, (iv) executes HyperCore actions via CoreWriter calls from the Pocket, and (v) executes registered protocol adapters by translating adapter `Call[]` into Pocket `execWithValue` / `approve` operations.
- **Pocket:** A minimal smart-wallet (clone) used by the Tailor. It can hold ERC20s and native tokens, transfer tokens, approve spenders, execute calls (with or without native value), and transfer native tokens out.
- **PocketFactory:** Deploys Pocket clones and tracks pockets created by the factory (via `isPocket`).
- **Adapters (MorphoBlueAdapter):** A pluggable integration mechanism where adapters generate and validate call bundles which the Tailor executes via the Pocket. `MorphoBlueAdapter` encodes the Morpho Blue interaction sequence (approval + Morpho call) so the Pocket supplies/withdraws collateral and borrows/repays against Morpho markets.

9 Attestation

Cyfrin attests that it performed a security review of the Button Update code provided by Button.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-02-13-cyfrin-button-update-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Button has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=069703faec51877cd45f6436e9f24de8f8d0b0051ee4325b622a6dc4fdd16dd3
F=2026-02-13-cyfrin-button-update-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.